

**BANARAS HINDU UNIVERSITY**  
B.Sc. (Hons.) Semester VI Examination, 2022-23

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**Zoology**

**Paper: ZOB-603 — Cell Biology and Genetics**

Time: Three hours

Full Marks: 70

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*Note: Answer three questions from Section-A (including Question No. 1 which is compulsory) and three questions from Section-B (including Question No. 1 which is compulsory). Use a separate answer book for each section. The figures in the right-hand margin indicate marks.*

**SECTION – A**  
**(Cell Biology) (Marks: 35)**

**Question 1.** Answer the following parts:

- (a) Define the importance of: [3 × 1 = 3]
- (i) Polyteny
  - (ii) Telomere
  - (iii) FG-repeats at nuclear pore complex.
- (b) Differentiate between the following: [2 × 2 = 4]
- (i) Tight junction and Gap junction
  - (ii) Integrins and Cadherins.
- (c) State giving reasons whether the following statements are True or False: [2 × 2 = 4]
- (i) Carrier transporters follow simple Michaelis-Menten kinetics of substrate transport.
  - (ii) Insulin receptors are an example of enzyme-coupled receptors.

**Question 2.** Answer the following:

- (a) What is meant by critical concentration for F-actin polymerization? Explain its role as treadmilling. [6]
- (b) Elaborate the different ways through which a cell can be transformed. [6]

**Question 3.** Answer the following:

- (a) Explain the organization of chromatin and its compaction into metaphase chromosome. [6]
- (b) How does SRP regulate co-translational protein transport across ER? In what way is the topology of the membrane-spanning polypeptide maintained during membrane insertion? [6]

**Question 4.** Answer the following:

- (a) What are G-proteins? Explain how G-protein signalling is regulated and activates second messengers. [6]
- (b) In what way can DNA damage freeze a potentially dividing cell population at the restriction point? [6]

**SECTION – B**  
**(Genetics) (Marks: 35)**

**Question 1.** Answer the following parts:

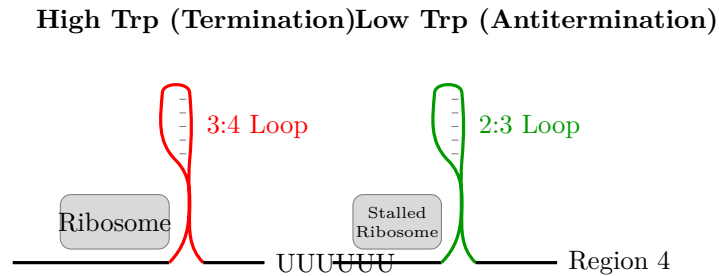
- (a) Write whether the following statements are True or False giving reasons in 2-3 sentences each: [3 × 2 = 6]
  - (i) Regulatory transcription factors bind to core promoter of a gene.
  - (ii) Proteome is larger than genome of a given species.
  - (iii) Tautomeric shifts within the bases that make up the nucleic acids are the most common cause of point mutations.
- (b) What is the status of Lac repressor and CAP protein under the following cellular conditions? [2 × 0.5 = 1]
  - (i) Lactose present and glucose absent
  - (ii) Lactose present and glucose present.
- (c) Distinguish between the following: [2 × 2 = 4]
  - (i) First division segregation and Second division segregation in *Neurospora*
  - (ii) Transposons and Retrotransposons.

**Question 2.** Answer the following:

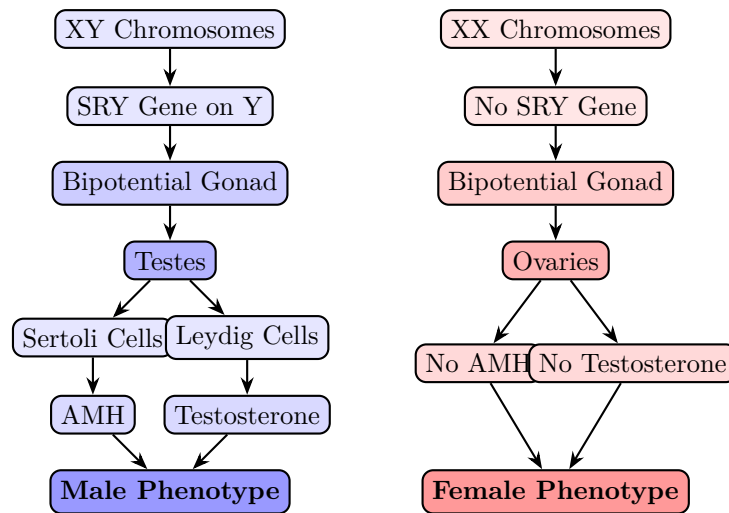
- (a) What is a 'three-point test cross'? Explain how genetic crosses are designed in *Drosophila* to derive a linkage map of three genes. [4]
- (b) Explain how gene regulation occurs via changes in DNA or chromatin structure. [4]
- (c) Describe in brief the 'general organisation of human genome'. [4]

**Question 3.** Answer the following:

- (a) Explain how mutations can be detected using the ClB technique in *Drosophila*. [4]  
 (b) How is 'trp' operon regulated by attenuation? [4]



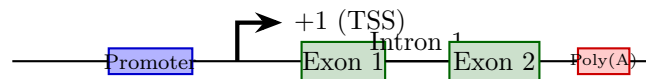
- (c) Describe with the help of a labelled diagram, the sex-determination pathway in human. [4]



**Question 4.** Write short notes on the following:

[4 × 3 = 12]

- (a) Ames test  
 (b) Eukaryotic transcription unit



- (c) Microarray technology  
 (d) Polygenic inheritance.

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